

SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT- 1

1. Student Details

Student Name	Homesh Sawarkar
PRN	202501110069
Division	A
Batch	A-2
Roll No.	202501110069
Date of Submission	09-09-2026

2. Practical Details

Practical No.	1
Practical Title	Class, Constructor, Destructor and User-Defined Functions
Date Performed	09-09-2026
Faculty Name	Dr. Khushal Khairnar

3. Problem Statement

Write a C++ program demonstrating the use of a class, a constructor, a destructor, and a user-defined member function. The program should create an object, initialize its data members using a constructor, display the stored student information using a user-defined function, and demonstrate object destruction using a destructor.

4. Objectives

1. To understand the concept and structure of a class and objects in C++.
2. To implement a constructor and destructor for object initialization and destruction.
3. To create and use a user-defined member function to perform an operation on object data.

5. Expected Learning Outcomes

- Understand the concept of classes and objects in C++.
- Learn how to use constructors and destructors.
- Understand and implement user-defined functions.
- Develop basic object-oriented programming skills.

6. Theory / Concept

A class is a blueprint used to create objects. A constructor automatically initializes the object when it is created. A destructor is automatically called when the object is destroyed. User-defined functions are created by the programmer to perform specific tasks, such as displaying student details and checking the result..

A destructor is a special member function written with a tilde (~) before the class name. It is automatically called when an object goes out of scope or is destroyed. It is used to perform cleanup operations.

A user-defined function is a function written by the programmer to perform a specific task. The displayDetails() member function displays the data stored in the Student object.

7. Algorithm / Pseudocode

START

- Define a Student class with private data members name, rollNo, and marks.
- Define a default constructor to initialize the student details.
- Define the user-defined display() function to display student details.
- Define the user-defined checkResult() function to check whether the student has passed or failed.
- Define a destructor to display a message when the object is destroyed.
- Create a Student object in main().
- Call display() and checkResult() using the object.
- End main(); the destructor is automatically invoked when the object goes out of scope.
- STOP

08. Technology / Tools / Libraries Used

Technology / Tool	Purpose
ProgrammingLanguage	C++
IDE / Platform	Visual Studio Code (VS Code)
Libraries / Packages	iostream, string

09. Implementation / GitHub Repository

Particular	Details
GitHub Repository URL	
Repository Name	Assignment 1 C++
Branch Used	main
Commit ID / Latest Commit	Add file via upload

Particular	Details
Date of Last Commit	09-09-2026

10. Test Cases / Input and Output

Test Case	Input	Expected Output	Actual Output	Status
1	Student s; Name = Homesh Roll No = 69 Marks = 87	Constructor message and Homesh's details with Pass result	Constructor message and Homesh's details with Pass result displayed	Pass
2	Student s; Name = Aryan Roll No = 24 Marks = 72	Constructor message and Aryan's details with Pass result	Constructor message and Aryan's details with Pass result displayed	Pass

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--- Student Details ---
Name : Homesh
Roll No: 69
Marks : 87
Result : Pass

Destructor called

```

11. Results / Observations

The C++ program was implemented successfully in Visual Studio Code (VS Code). The Student class was created with private data members. The parameterized constructor initialized the object, the user-defined displayDetails() function displayed the student information, and the destructor was automatically called when the object was destroyed.

12. Conclusion

The practical successfully demonstrated the implementation of a class, object, constructor, destructor, and user-defined member function in C++. It helped in understanding object initialization, member-function invocation, and automatic object destruction in object-oriented programming.

13. Assignment Questions

Q1. What is a constructor in C++? Explain its purpose.

Answer:

A constructor is a special member function of a class that has the same name as the class. It is automatically called when an object is created and is mainly used to initialize the data members of the object. In this program, the parameterized Student constructor initializes name, rollNo, and marks.

Q2. What is the difference between a constructor and a destructor?

Answer:

A constructor is called automatically when an object is created and is used for initialization. A destructor is called automatically when an object is destroyed or goes out of scope and is used for cleanup. A constructor has the class name, while a destructor has the class name preceded by a tilde (~). A class can have constructors, including overloaded constructors, whereas it has only one destructor.

14. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name

	PRN	Signature
Homesh Sawarkar	202501110069	Homesh.A.S